REAL ANALYSIS 2

SPRING 2015

JAMES ROSADO

PROFESSOR OSLER

A Study of Two Convergence Theorems

Part 1: An Explanation of the *Monotone Convergence Theorem*

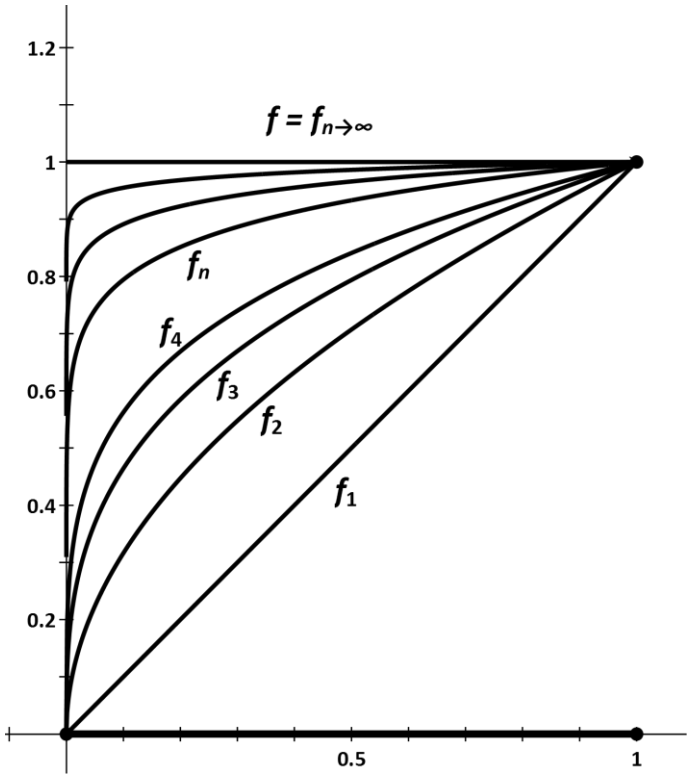
Theorem 28.2 from "An Introduction to Lebesgue Integration and Fourier Series" states the following:

> Let *A* be a bounded measurable subset of $\mathbb{R}$ and $\{f_n\}$ is a sequence of measurable functions on *A* such that $0 \le f_1 \le f_2 \le \cdots$. Let *f* be the (pointwise) limit of $\{f_n\}$. Then *f* is integrable and $\lim_{n\to\infty} \int_A f_n dm = \int_A f \, dm$

If we analyze each line of this theorem, the first line is describing a sequence of functions with each function less than or equal to the next one. As an example let us consider a the following function on the domain $A = [0,1] \subset \mathbb{R}$: $f_n(x) = x^{1/n} \quad n = 1, 2, 3....$

Below are the graphs for some values of *n*. Notice that $0 \le x \le x^{1/2} \le x^{1/3} \le \cdots \le 1$; in fact, our function is bounded above by 1 and below by 0. If we take the limit $n \to 0$:

$$\lim_{n\to\infty} x^{1/n} = x^{\left(\lim_{n\to\infty} 1/n\right)} = x^{\,0} = 1$$



Our set is measurable, $m(A) = 1$. Now our example function is measurable, if we consider the basic definition of a measurable function, *f* is measurable if $\{x \in A \mid c < f(x)\}$ is measurable for all *c*. There are three possible cases for our particular example,

$c \in (-\infty, 0]$	$m\left(\{x \in A \mid c < f(x)\}\right) = 1$
$c \in (0,1)$	$m\left(\{x \in A \mid c < f(x)\}\right) = c^n$
$c \in [1, \infty)$	$m\left(\{x \in A \mid c < f(x)\}\right) = 0$

Notice that for each case *f* is measurable. The next part of the theorem states that *f* is the limit and pointwise limit of f_n. What this means is that you pick a particular *x* and plug it in to each f_n you will eventually converge to the same value if you plugged *x* directly into *f*. Notice in the table below for an example *x* = 0.5:

n	$f_n(x)$
1	0.5
2	0.70710678…
3	0.79370052…
4	0.84089641…
5	0.87055056…
6	0.89089871…
7	0.90572366…
9	0.92587472…
90	0.99232794…
900	0.99923013…
9000	0.99992298…

Notice that the set $\{f_1(0.5), f_2(0.5), f_3(0.5)... f_n(0.5)...\} \underset{n\to\infty}{\to} f(0.5) = 1$, and we can repeat this for every *x* in [0,1] therefore, *f* is the pointwise limit of $\{f_n\}$. The conclusion of the theorem states that the function *f* is therefore integrable; in fact, if we integrate the sequence of functions $\{f_n\}$ it will converge to the same integral of *f*. So if we consider our example, and integrate each iteration on *n*:

n	f_n	$\int f_n(x)\,dx$
1	x	$\frac{1}{2}x^2 + C$
2	$x^{1/2}$	$\frac{3}{2}x \cdot x^{1/2} + C$
3	$x^{1/3}$	$\frac{4}{3}x \cdot x^{1/3} + C$
4	$x^{1/4}$	$\frac{5}{4}x \cdot x^{1/4} + C$
5	$x^{1/5}$	$\frac{6}{5}x \cdot x^{1/5} + C$
6	$x^{1/6}$	$\frac{7}{6}x \cdot x^{1/6} + C$
7	$x^{1/7}$	$\frac{8}{7}x \cdot x^{1/7} + C$

8	$x^{1/8}$	$\frac{9}{8}x\cdot x^{1/8}+C$

Notice that the pattern is:

$$\int f_n(x)\,dx = \tfrac{n+1}{n}x\cdot x^{1/n}$$

If we take the limit $n\to\infty$ in the above line we get,

$$\begin{aligned}\lim_{n\to\infty}\int f_n(x)\,dx &= \lim_{n\to\infty}\tfrac{n+1}{n}x\cdot x^{1/n}\\ &= x\lim_{n\to\infty}\tfrac{n+1}{n}x^{1/n}\\ &\to x\cdot\left(1\cdot x^{0}\right)=x\end{aligned}$$

Notice also that the integral of $f(x)=1$ is equal to the same result as above "*x*."

PART 2: Explanation of the Proof.

First notice that we are given a sequence of measurable functions $\{f_n\}$ and *f* is the point wise limit of this sequence; therefore, by Theorem 18.6 on page 52:

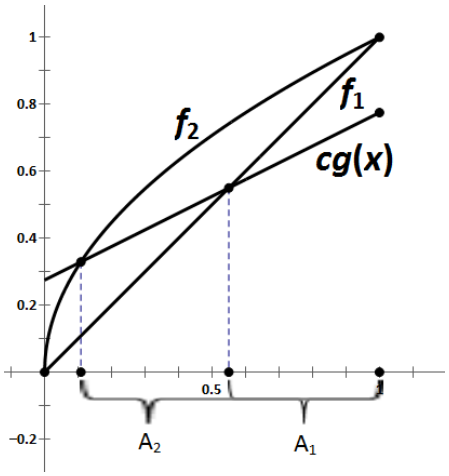
> If $f_n : A\to\mathbb{R}$ is measurable for each $n=1,2,3,\ldots$ and *f* is the point wise limit of $\{f_n\}$, then *f* is measurable on *A*.

If *f* is measurable then it must be integrable and $f\ge 0$. Since $0\le f_1\le f_2\le\cdots\le f_n\cdots\le f$, then the integrals under each curve f_n will be less than the integral of the next *n* + 1 curve f_{n+1}, hence

$$\int_A f_1\,dm\le\int_A f_2\,dm\le\int_A f_3\,dm\le\cdots\le\int_A f_n\,dm\le L\le\int_A f\,dm$$

Now we need to show that *L* is the limit of the integral. So we will consider a value *c* such that $0 < c < 1$ and a function *g* such that $0\le g\le f$. I will also use the diagram for reference. Notice in the diagram to the right

$$A_1=\{x\mid f_1(x)\ge cg(x)\}$$
$$A_2=\{x\mid f_2(x)\ge cg(x)\}$$



And that $A_1 \subset A_2$. If we extend this for more f_n we get that $A_1 \subset A_2 \subset A_3 \subset \cdots A_n \subset A$ and that $A = \bigcup_{k=1}^{\infty} A_k$ with all subsets measurable on *A*. Since $f_n \geq cg$ over these subsets, that means the integrals must be greater or equal,

$$\int_{A_n} f_n \, dm \geq c \int_{A_n} g \, dm$$

Keep in mind that $A_1 \subset A_2 \subset A_3 \subset \cdots A_n \subset A$, so the integral over the unions of all the A_n's will have the greatest value compared to each subset only, hence,

$$\int_{\cup A_n} f_n \, dm \geq \int_{A_n} f_n \, dm \geq c \int_{A_n} g \, dm$$

If we let $n \to \infty$ then we get our limiting value:

$$L = \lim_{n \to \infty} \int_{\cup A_n} f_n \, dm = \lim_{n \to \infty} \int_{A} f_n \, dm \geq \int_{A_n} f_n \, dm \geq c \int_{A_n} g \, dm$$

This implies then that,

$$L \geq c \int_{A_n} g \, dm$$

But recall our previous assertion if *L* is greatest on just one of the subsets of *A*, then it must be greatest on the union of all the subsets (over all of *A*):

$$L \geq c \int_{A} g \, dm$$

This is valid for all values of *c* in [0,1], so this puts a least upper bound of our integral:

$$L \geq \int_{A} f \, dm$$

Recall, that is greater than *L* from earlier,

$$L \geq \int_{A} f \, dm \geq L$$

So this integral must be "squeezed" to this *L* and this is also the limit of the integrals for f_n, this in turn means that the limit of the f_n's integral is equal to *L* which is equal to the integral on *f* itself or:

$$\lim_{n\to\infty}\int_A f_n\ dm = L = \int_A f\ dm$$

PART 3: Example 1 of the Monotone Convergence Theorem.

Recall our sequence of functions $\left\{ f_n = x^{1/n} \mid n = 1,2,3... \right\} = \left\{ x, x^{1/2}, x^{1/3}... \right\}$ over $A = [0,1] \subset \mathbb{R}$ and $f_n : A \to \mathbb{R}$. Let us consider the left hand side of the theorem:

$$\int_A f_n\ dm = \int_{[0,1]} x^{1/n} dm$$

We will consider the Lebesgue sum of this integral:

$$\begin{aligned}\int_{[0,1]} x^{1/n} dm &= \lim_{\|P\|\to 0} \sum_{k=1}^{N} y_k^* \cdot m\left(\left\{ x \in A \mid y_k \le x^{1/n} < y_{k+1} \right\} \right) \\ &= \lim_{\|P\|\to 0} \sum_{k=1}^{N} y_k^* \cdot m\left(A_k \right)\end{aligned}$$

But if we take the limit $n \to \infty$ then $y_k^* \to 1$ because $\lim_{n\to\infty} x^{1/n} = 1$;therefore, our Lebesgue sum becomes:

$$\lim_{\|P\|\to 0} \sum_{k=1}^{N} m\left(A_k \right) = \lim_{\|P\|\to 0} m\left(\bigcup_{k=1}^{N} A_k \right)$$

As the partitions get smaller and smaller and we keep adding more along the y-axis we start filling the domain *A* with more and more subsets A_k. Hence,

$$\lim_{\|P\|\to 0} \sum_{k=1}^{N} m\left(A_k \right) = \lim_{\|P\|\to 0} m\left(\bigcup_{k=1}^{N} A_k \right) = m(A) = 1$$

This means that the left hand of our integral taken to the limit $n \to \infty$ is,

$$\lim_{n\to\infty} \int_{[0,1]} x^{1/n} dm = 1$$

Let us now verify the right hand side, we know that $f = 1$ is the pointwise limit of f_n as $n \to \infty$,

$$\int_A f\ dm = \int_{[0,1]} 1 \cdot dm = 1$$

Hence we have arrived at an example of our theorem,

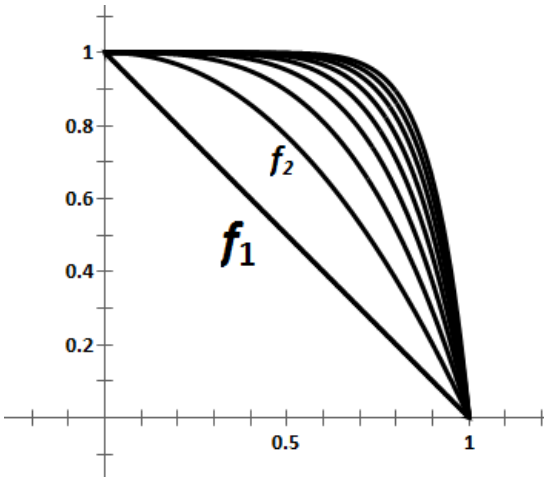
$$\lim_{n\to\infty}\int_{[0,1]} x^{1/n}dm = 1 = \int_{[0,1]} 1\cdot dm$$

PART 4: Example 2 of the Monotone Convergence Theorem.

Let us consider the sequence of functions defined on $A=[0,1]\subset\mathbb{R}$, $\{f_n = 1-x^n \mid n=1,2,3...\}$ and as *n* increases we have,

$$0\le 1-x\le 1-x^2\le 1-x^3\le\cdots\le 1-x^n$$

It is important to note that we are considering the interval [0,1] so as $n\to\infty$ then $1-x^n\to 1$. Below is a graph for a few iterations of *n*.



Let us now consider the integral below,

$$\begin{aligned}\int_{[0,1]} 1-x^n dm &= \int_{[0,1]} 1\cdot dm - \int_{[0,1]} x^n dm\\ &= 1-\int_{[0,1]} x^n dm\\ &= 1-\lim_{\|P\|\to 0} L(x^n,P)\end{aligned}$$

Notice that in the last equality above we utilize the Lebesgue sum and allow the partitions to progressively get smaller and smaller. Since we are integrating over [0,1] that means *x* will be a decimal less than 1, but as we raise *x* to the power of *n* and let $n\to\infty$ all these *x*'s will tend to 0. Therefore,

$$\begin{aligned}\lim_{n\to\infty}\int_{[0,1]} 1-x^n dm &= \lim_{n\to\infty}\left(1-\lim_{\|P\|\to 0} L(x^n,P)\right)\\ &= 1-\lim_{\|P\|\to 0} L(0,P)\\ &= 1-0\end{aligned}$$

$$\lim_{n\to\infty}\int_{[0,1]} 1-x^n dm = 1$$

Recall that $n \to \infty$ means $1 - x^n \to 1$ on [0,1] and we get the same integral as in the previous example:

$$\int_A f \, dm = \int_{[0,1]} 1 \cdot dm = 1$$

And we arrive at the same result,

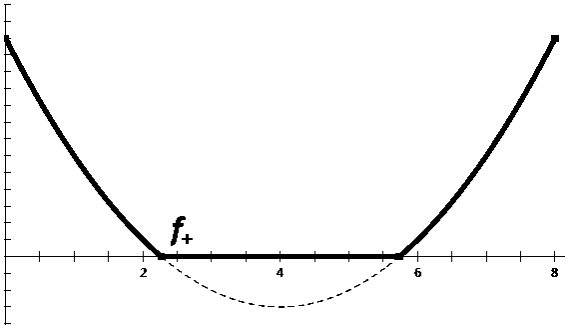
$$\lim_{n\to\infty} \int_{[0,1]} 1 - x^n dm = 1 = \int_{[0,1]} 1 \cdot dm$$

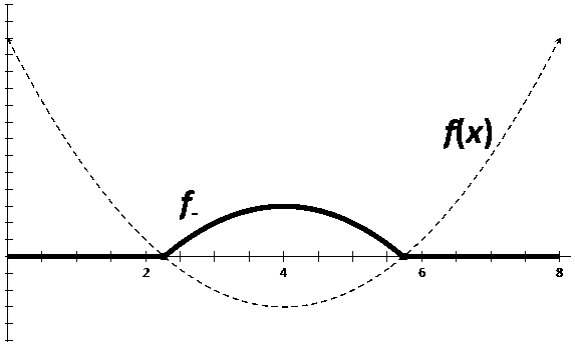
PART 5: An Explanation of the *Lebesgue Dominated Convergence Theorem*

Theorem 28.9 from "An Introduction to Lebesgue Integration and Fourier Series" states the following:

> Let *A* be a bounded measurable subset of $\mathbb{R}$ and $\{f_n\}$ is a sequence of measurable functions on *A* such that $\lim_{n\to\infty} f_n(x) = f(x)$ for every $x \in A$. If there exists a function $g \in \mathcal{L}$(A) [i.e. *g* is summable] such that $|f_n(x)| \le g(x)$ for *n* = 1,2,3… and all $x \in A$, then $\lim_{n\to\infty} \int_A f_n dm = \int_A f \, dm$.

The first sentence that the limit of a function that depends on *n* is the function *f.* So consider our first example $\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} x^{1/n} = 1 = f(x)$, as you can see the limit approaches *f,* notice that each f_n is measurable on *A* and the limit exists for every $x \in A$. The second sentence refers to the existence of some function *g* (a dominating function) that is in $\mathcal{L}$(A). The $\mathcal{L}$(A) is a set of functions that map $A \to \mathbb{R}$ and each function is summable. The idea of summability is that a function can be though up as a "positive" part and a "negative" part that are linearly combined, if the said parts both have finite integrals then the function is summable. As an example of this "summability" let us consider the function $f(x) = (x-4)^2 - 3$.





Above are examples of these positive and negative parts, notice that $f(x) = f_+(x) - f_-(x)$. If we integrate the two above functions over the domain [0,8] we will get finite values for the integrals, this can be thought of as the "area" under the curves for this particular example:

$$M \geq \int_A f_+ \, dm \geq 0 \text{ and } M \geq \int_A f_- \, dm \geq 0$$

This also implies that $\int_A f \, dm = \int_A f_+ \, dm - \int_A f_- \, dm$. Now that we have established this idea of summability of a function, if we consider the last part of the theorem it states that this function *g* is always greater or equal to the magnitude of our sequence of functions; in other words, our sequence of functions is always bounded in size. As an example let us consider the sequence of function

$$\left\{ f_n(x) = \frac{1}{x}\cos\left(\frac{x}{n}\right) | \, n = 1, 2, 3... \right\}$$

Notice that if we let $n \to \infty$, then $f(x) = \frac{1}{x}$ and $| f_n | \leq f$; therefore, our f is this bounding function. The only issue is that our functions are not summable at the origin, but if consider other intervals not near the origin say $A = [\pi, 2\pi]$ then our integrals are finite. If these criteria are met a) summability and b) boundedness then we arrive at the same conclusion as we did with the Monotone Convergence Theorem, except our sequence of functions does not have to be monotonic, i.e. $\left\{ f_n(x) = \frac{1}{x}\cos\left(\frac{x}{n}\right) | \, n = 1, 2, 3... \right\}$ does not have the property of $0 \leq f_1 \leq f_2 \leq \cdots$, monotonicity.

PART 6: Proof of the *Lebesgue Dominated Convergence Theorem*

To prove this theorem we need to utilize Fatou's Lemma which states that given a sequence of non-negative measurable functions on *A*, and $f(x) = \inf \{ f_n(x) | \, n = 1, 2, 3... \}$ for every $x \in A$, then $\int_A f \, dm \leq \inf \left\{ \int_A f_n \, dm | \, n = 1, 2, 3... \right\}$. Now let us get into the proof, first notice that since *f* and f_n are measurable (and integrable) then that means they are both summable and have finite integrals, $f \in \mathcal{L}$(A) and $f_n \in \mathcal{L}$(A). Let us not forget that there exists another function $g \in \mathcal{L}$(A) such that $|f_n(x)| \leq g(x)$.

Therefore, $f_n + g \geq 0$ and by Fatou's Lemma we can conclude that $\int_A f + g \, dm \leq \inf \left\{ \int_A f_n + g \, dm | \, n = 1, 2, 3... \right\}$. Notice that the integrand of *g* is on either side of the inequality; hence,

(1) $$\int_A f \, dm \leq \inf \left\{ \int_A f_n \, dm | \, n = 1, 2, 3... \right\}.$$

Recall that we are dealing with $|f_n(x)|$ so f_n maybe positive or negative; with that $g - f_n \geq 0$ and utilizing Fatou's Lemma we arrive at, $\int_A g - f \; dm \leq \inf\left\{\int_A g - f_n \; dm \mid n = 1,2,3...\right\}$ and the a similar conclusion as before $\int_A -f \; dm \leq \inf\left\{\int_A -f_n \; dm \mid n = 1,2,3...\right\}$. If we divide out the negation we obtain,

(2) $$\int_A f \; dm \geq \sup\left\{\int_A f_n \; dm \mid n = 1,2,3...\right\}$$

Combining (1) and (2) we get,

$$\int_A f \; dm \leq \inf\left\{\int_A f_n \; dm \mid n = 1,2,3...\right\} \leq \sup\left\{\int_A f_n \; dm \mid n = 1,2,3...\right\} \leq \int_A f \; dm$$

Hence as $n \to \infty$, then the middle of the above inequality equal the outer part of the inequality and $\lim_{n\to\infty}\int_A f_n dm = \int_A f \; dm$.

PART 7: Example 1 of the Lebesgue Dominated Convergence Theorem.

In this example I will show that the function defined as,

(3) $$F(t) = \int_{[1,\infty)} \frac{1}{x^2}\sin(\pi t)\, dm$$

is continuous. First, let us establish a dominating function *g*, let,

(4) $$g(x) = \chi_{[1,\infty)}(x) \cdot \frac{1}{x^2} = \begin{cases} 1/x^2 & x \geq 1 \\ 0 & x < 1 \end{cases}$$

But we need to show that *g* is summable, this means that the integral of *g* on [1,∞) is finite. Notice $\frac{1}{x^2}$ is continuous on [1,∞) and $\chi_{[1,\infty)}(x)$ is measurable; therefore, *g* is measurable. If *g* is summable, then $\int_{[1,\infty)} g \; dm < \infty$. So let us consider the limit problem,

$$\int_{[1,\infty)} g \; dm = \lim_{n\to\infty}\int_{[1,\infty)} \chi_{[1,n)}(x) \cdot \frac{1}{x^2} dm$$

$$= \lim_{n\to\infty}\int_{[1,n)} \frac{1}{x^2} dm$$

$$\int_{[1,\infty)} g \; dm = 1 < \infty$$

So *g* is summable. If *F*(*t*) is continuous then,

$$\lim_{n\to\infty}\int_{[1,n)}\frac{1}{x^2}\sin(\pi t)\,dm=\int_{[1,\infty)}\frac{1}{x^2}\sin(\pi t)\,dm$$

We have found our dominating function and $g(x)=\chi_{[1,\infty)}(x)\frac{1}{x^2}$ and we let $f(x)=\chi_{[1,\infty)}(x)\frac{1}{x^2}\sin(\pi t)$, then $\left|\chi_{[1,\infty)}(x)\frac{1}{x^2}\sin(\pi t)\right|\le\chi_{[1,\infty)}(x)\frac{1}{x^2}$; therefore,

$$\lim_{n\to\infty}\int_{[1,n)}\frac{1}{x^2}\sin(\pi t)\,dm=\int_{[1,\infty)}\frac{1}{x^2}\sin(\pi t)\,dm$$

PART 8: Example 2 of the Lebesgue Dominated Convergence Theorem.

Let us consider another example let $f_n(x)=\frac{1-nx^n}{x+1}$ on the interval $A=[0,1]$ and the limit as $n\to\infty$ will yield $f=\frac{1}{1+x}$. Let $g=\frac{1}{1+x}$ and notice that $|f_n(x)|\le g(x)$; hence, *g* is our dominating function that f_n approaches and *g* is summable:

$$\left|\frac{1-nx^n}{1+x}\right|\le\frac{1}{1+x}$$

Therefore,

$$\begin{aligned}\lim_{n\to\infty}\int_{[0,1]}\frac{1-nx^n}{1+x}dm&=\int_{[0,1]}\frac{dm}{1+x}\\&=\int_0^1\frac{dx}{1+x}\\&=\ln(1+\mathrm{x})\Big|_0^1=\ln(2)\end{aligned}$$